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$$I = \int x \ln x dx$$

$$\left\{ \begin{array}{l} u = \ln x \\ dv = x dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = \frac{1}{x} dx \\ v = \frac{x^2}{2} \end{array} \right.$$

$$I = \ln x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} dx$$

$$= \frac{1}{2} x^2 \ln x - \int \frac{x^2}{2x} dx$$

$$= \frac{1}{2} x^2 \ln x - \frac{1}{2} \int x dx$$

$$= \frac{1}{2} x^2 \ln x - \frac{x^2}{4} + c$$

References